



Grand Polo Club & Resort SOUND SYSTEM PROFILE

DETAILED DESIGN

DATE: 21-01-2026

Quality Information

Document Grand Polo Club & Resort

Ref

Date 21-01-2026

Prepared by Musthafa Machingal

Revision History

Revision	Revision Date	Details	Authorised	
			Name/Position	Signature
00	21-01-2026	Sound System Profile	Musthafa Machingal	

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1.0 Introduction

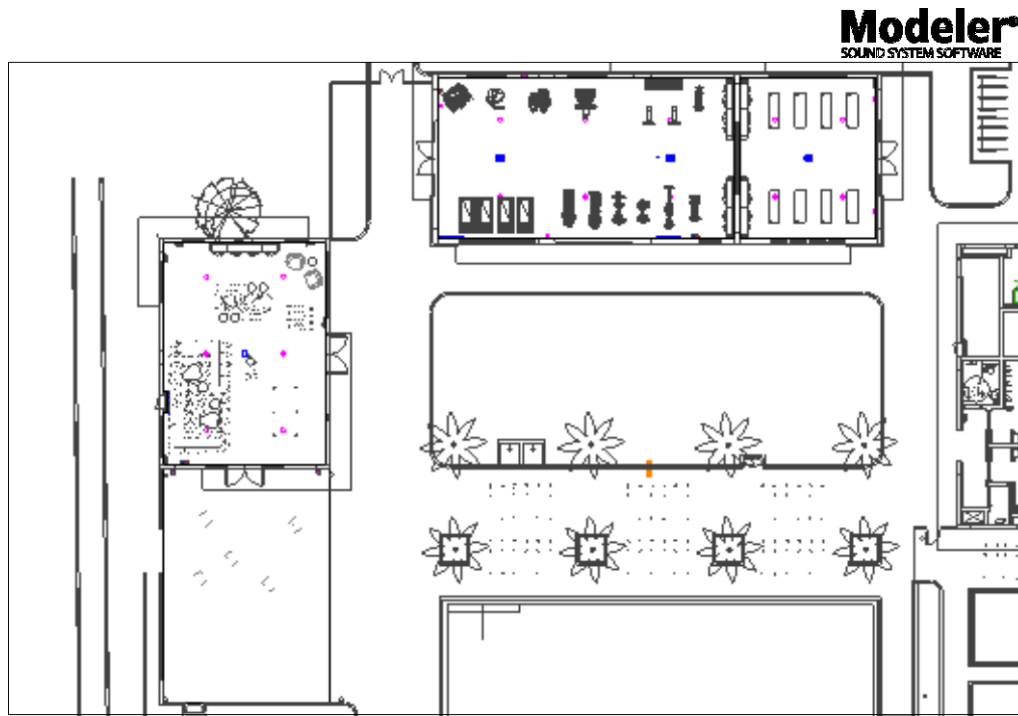
Grand Polo Club & Resort is an upcoming project in United Arab Emirates. A state of art installed sound reinforcement system is being proposed by BOSE PROFESSIONAL for the Background Music.

This report shall provide an overall system profile, description and expected Direct Sound Coverage levels of the new proposed sound system using BOSE Professional grade loudspeakers.

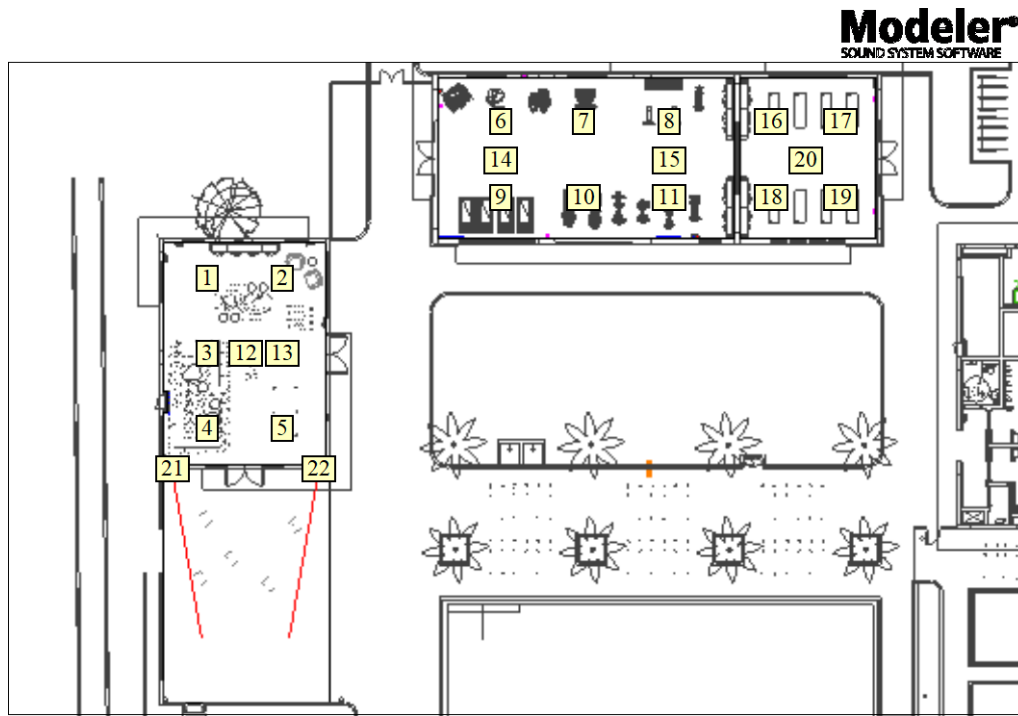
2.0 3D Modelling (COMMUNITY CENTER TH-01)

2.1 Electro-Acoustical room model

Based on the architectural drawings of the venue, we created a new electro-acoustical room model.



3.0 Loudspeaker Layout



3.1 Loudspeaker Database

Details																
	Cluster	Zone	Tier	Index	Description	Speaker	Status	Solo	X	Y	Z	Roll°	Pitch°	Yaw°	Pwr	Tap
1	1	1	1	1	▼	Bose DM3C 100V	On	☐	-26.000	11.200	3.000	0.0	-90.0	0.0	25	▼
2	2	1	1	1	▼	Bose DM3C 100V	On	☐	-21.500	11.200	3.000	0.0	-90.0	0.0	25	▼
3	3	1	1	1	▼	Bose DM3C 100V	On	☐	-26.000	6.700	3.000	0.0	-90.0	0.0	25	▼
4	4	1	1	1	▼	Bose DM3C 100V	On	☐	-26.000	2.200	3.000	0.0	-90.0	0.0	25	▼
5	5	1	1	1	▼	Bose DM3C 100V	On	☐	-21.500	2.200	3.000	0.0	-90.0	0.0	25	▼
6	6	1	1	1	▼	Bose DM3C 100V	On	☐	-8.750	20.400	5.000	0.0	-90.0	0.0	25	▼
7	7	1	1	1	▼	Bose DM3C 100V	On	☐	-3.750	20.400	5.000	0.0	-90.0	0.0	25	▼
8	8	1	1	1	▼	Bose DM3C 100V	On	☐	1.250	20.400	5.000	0.0	-90.0	0.0	25	▼
9	9	1	1	1	▼	Bose DM3C 100V	On	☐	-8.750	15.900	5.000	0.0	-90.0	0.0	25	▼
10	10	1	1	1	▼	Bose DM3C 100V	On	☐	-3.750	15.900	5.000	0.0	-90.0	0.0	25	▼
11	11	1	1	1	▼	Bose DM3C 100V	On	☐	1.250	15.900	5.000	0.0	-90.0	0.0	25	▼
12	12	1	1	1	▼	Bose DM8C-SUB 100V	On	☐	-23.700	6.700	3.000	0.0	-90.0	0.0	150	▼
13	13	1	1	1	▼	Bose DM3C 100V	On	☐	-21.500	6.700	3.000	0.0	-90.0	0.0	25	▼
14	14	1	1	1	▼	Bose DM8C-SUB 100V	On	☐	-8.700	18.100	5.000	0.0	-90.0	0.0	150	▼
15	15	1	1	1	▼	Bose DM8C-SUB 100V	On	☐	1.300	18.100	5.000	0.0	-90.0	0.0	150	▼
16	16	1	1	1	▼	Bose DM3C 100V	On	☐	7.350	20.400	3.000	0.0	-90.0	0.0	25	▼
17	17	1	1	1	▼	Bose DM3C 100V	On	☐	11.350	20.400	3.000	0.0	-90.0	0.0	25	▼
18	18	1	1	1	▼	Bose DM3C 100V	On	☐	7.350	15.900	3.000	0.0	-90.0	0.0	25	▼
19	19	1	1	1	▼	Bose DM3C 100V	On	☐	11.350	15.900	3.000	0.0	-90.0	0.0	25	▼
20	20	1	1	1	▼	Bose DM8C-SUB 100V	On	☐	9.400	18.100	3.000	0.0	-90.0	0.0	150	▼
21	21	1	1	1	▼	Bose DM3SE 100V	On	☐	-28.000	-0.200	4.000	90.0	0.0	-80.0	25	▼
22	22	1	1	1	▼	Bose DM3SE 100V	On	☐	-19.400	-0.200	4.000	90.0	0.0	-100.0	25	▼

4.0 System Performance

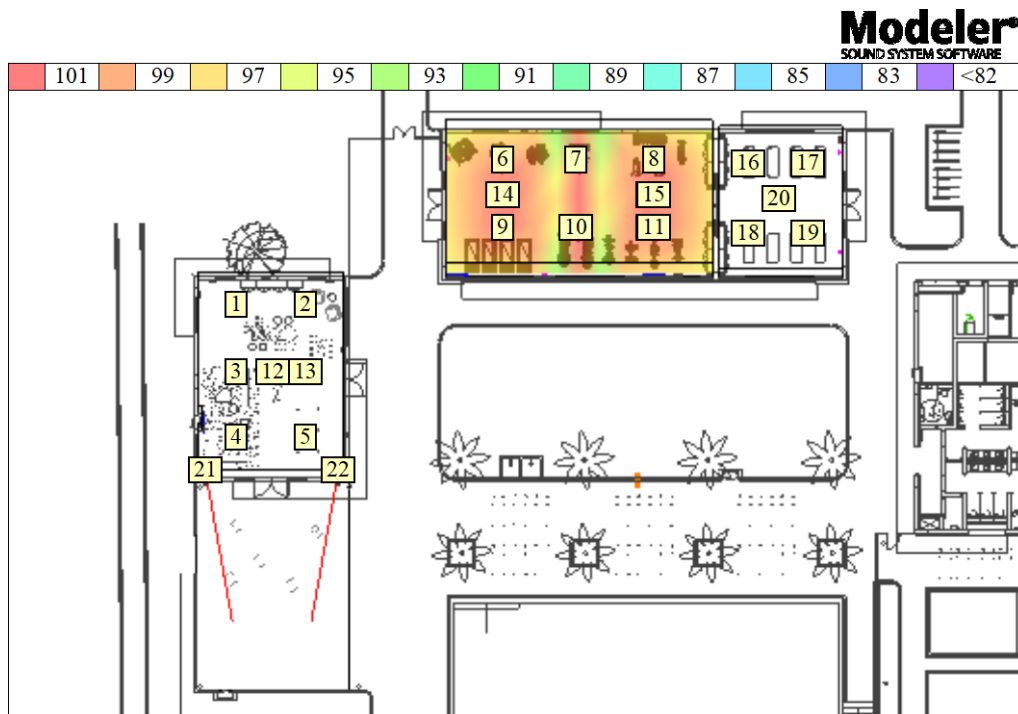
4.1 Calculation Assumptions

- Power source: Broadband Pink Noise (6dB CF)
- Power settings (Loudspeakers): Maximum continuous (RMS) power ()
- Direct field coverage (Broadband, dBZ): Direct sound levels
- Listening Height: 1.5m from FFL

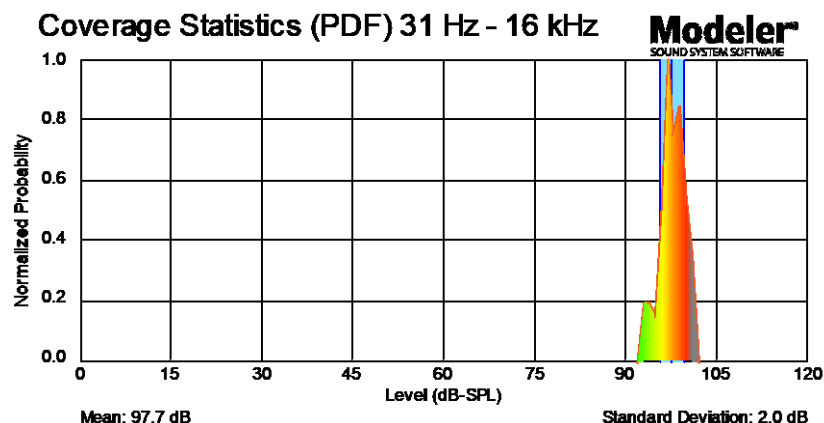
4.2 Direct field Coverage (31-16kHz) (GYM)

The first goal in creating a good sound system is to provide even direct sound intensity over the entire audience area and avoiding direct sound coverage in the non-occupied areas of the venue. The direct field is the sound that reaches the listener's ears directly from the speaker(s) in the design and it does not consider any sound reflections.

The coverage map/s below shows the direct field coverage for the venue. Good direct field coverage appears as relatively evenly coloured coverage area, without distinct dropout spots. This ensures even coverage of sound intensity over the coverage areas.



The information of this direct field (31-16kHz) coverage map can be condensed to a single figure of merit. The graph containing this figure, called a histogram, shows how often a given sound pressure level occurs in the model. Below you can find the SPL histogram of the direct field coverage between 31-16kHz. The frequency of occurrence (normalized) of a particular SPL is seen on the Y-axis of the graph while the X-axis shows the SPL in decibels.



A room with ideal coverage would have the same SPL at any point in the room and would appear as a single spike at that SPL level axis. As coverage becomes less than ideal the curve widens out to include other SPL levels.

The coverage data is processed in Modeler to yield the mean SPL level and the standard deviation. For the purposes of determining the quality or evenness of the coverage, it is the standard deviation that is important. (Applying more power to the system will increase the mean SPL level but the coverage pattern will be the same.)

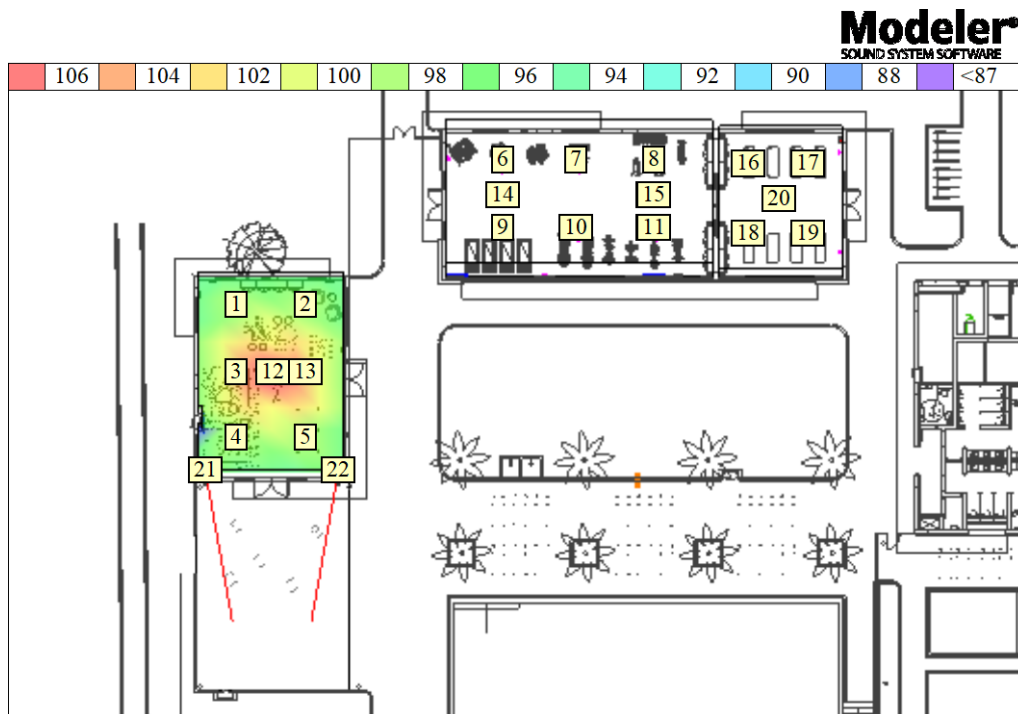
Referring to the standard practice of statistics, 68% of the samples are within $\pm 1\text{StDv}$ and 95% of the samples are within $\pm 2\text{StDv}$.

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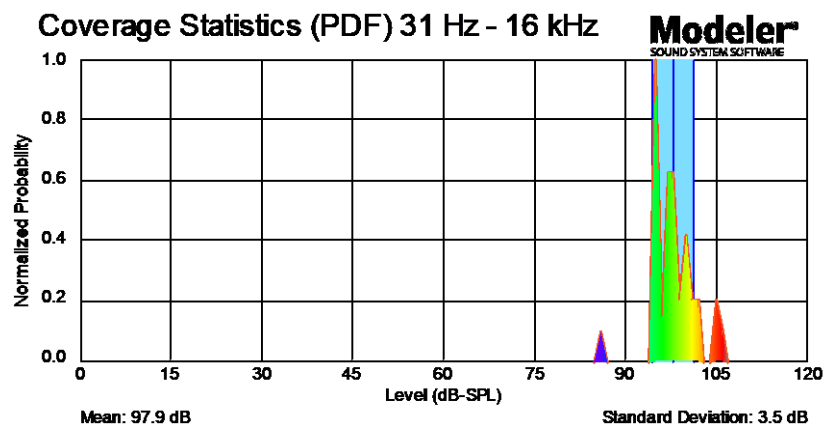
4.3 Direct field Coverage (31-16kHz) (COMMUNITY HALL)

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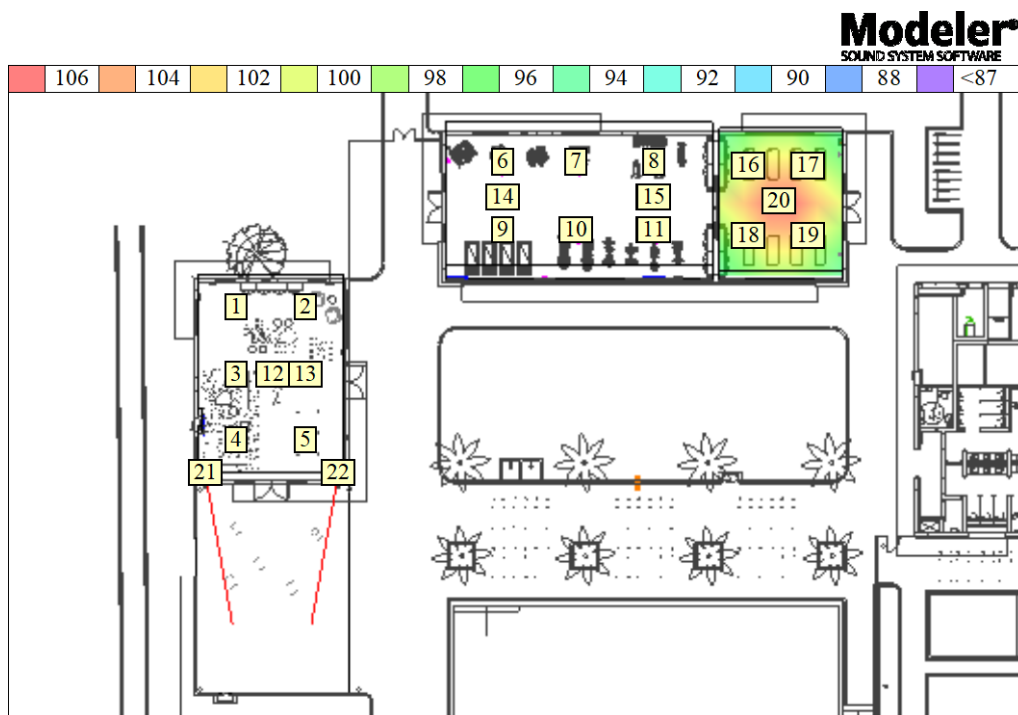
The coverage data is processed in Modeler to yield the mean SPL level and the standard deviation. For the purposes of determining the quality or evenness of the coverage, it is the standard deviation that is important. (Applying more power to the system will increase the mean SPL level but the coverage pattern will be the same.)

Referring to the standard practice of statistics, 68% of the samples are within +/- 1StDv and 95% of the samples are within +/- 2StDv.

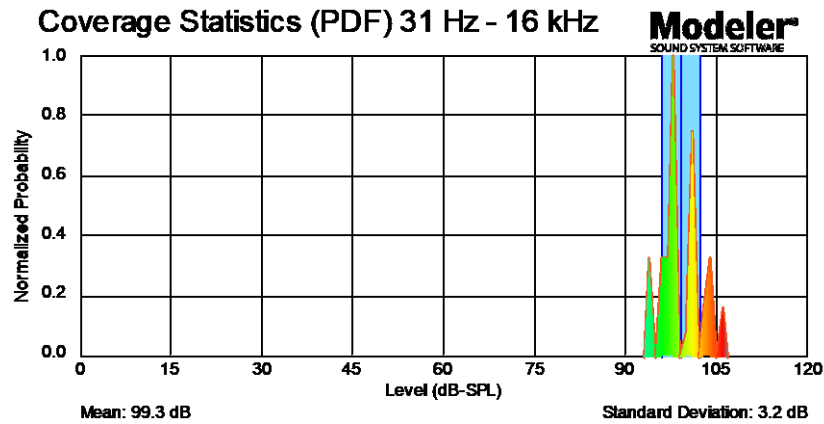
4.4 Direct field Coverage (31-16kHz) (YOGA)

The first goal in creating a good sound system is to provide even direct sound intensity over the entire audience area and avoiding direct sound coverage in the non-occupied areas of the venue. The direct field is the sound that reaches the listener's ears directly from the speaker(s) in the design and it does not consider any sound reflections.

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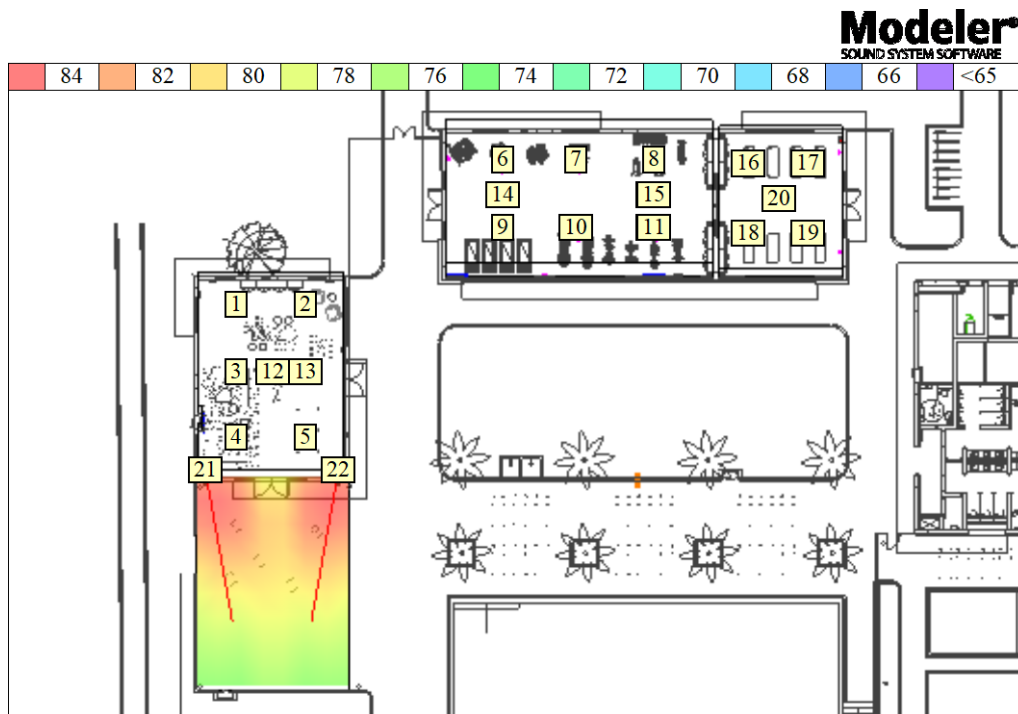
The coverage data is processed in Modeler to yield the mean SPL level and the standard deviation. For the purposes of determining the quality or evenness of the coverage, it is the standard deviation that is important. (Applying more power to the system will increase the mean SPL level but the coverage pattern will be the same.)

Referring to the standard practice of statistics, 68% of the samples are within $\pm 1\text{StDv}$ and 95% of the samples are within $\pm 2\text{StDv}$.

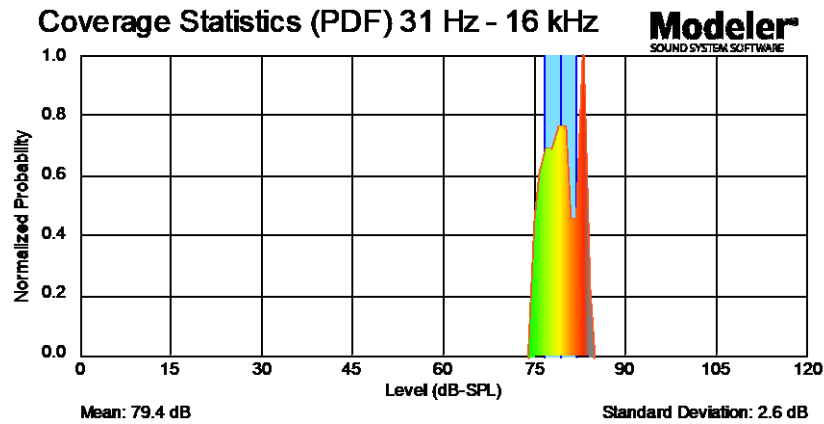
4.5 Direct field Coverage (31-16kHz) (GAMES AREA)

The first goal in creating a good sound system is to provide even direct sound intensity over the entire audience area and avoiding direct sound coverage in the non-occupied areas of the venue. The direct field is the sound that reaches the listener's ears directly from the speaker(s) in the design and it does not consider any sound reflections.

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A room with ideal coverage would have the same SPL at any point in the room and would appear as a single spike at that SPL level axis. As coverage becomes less than ideal the curve widens out to include other SPL levels.

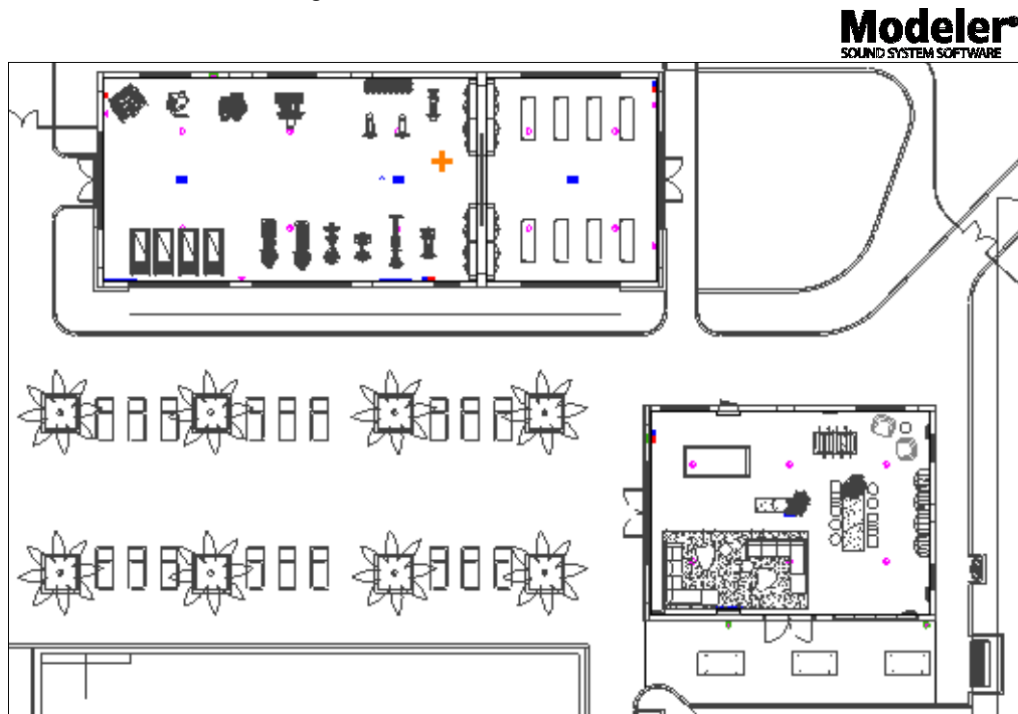
The coverage data is processed in Modeler to yield the mean SPL level and the standard deviation. For the purposes of determining the quality or evenness of the coverage, it is the standard deviation that is important. (Applying more power to the system will increase the mean SPL level but the coverage pattern will be the same.)

Referring to the standard practice of statistics, 68% of the samples are within $\pm 1\text{StDv}$ and 95% of the samples are within $\pm 2\text{StDv}$.

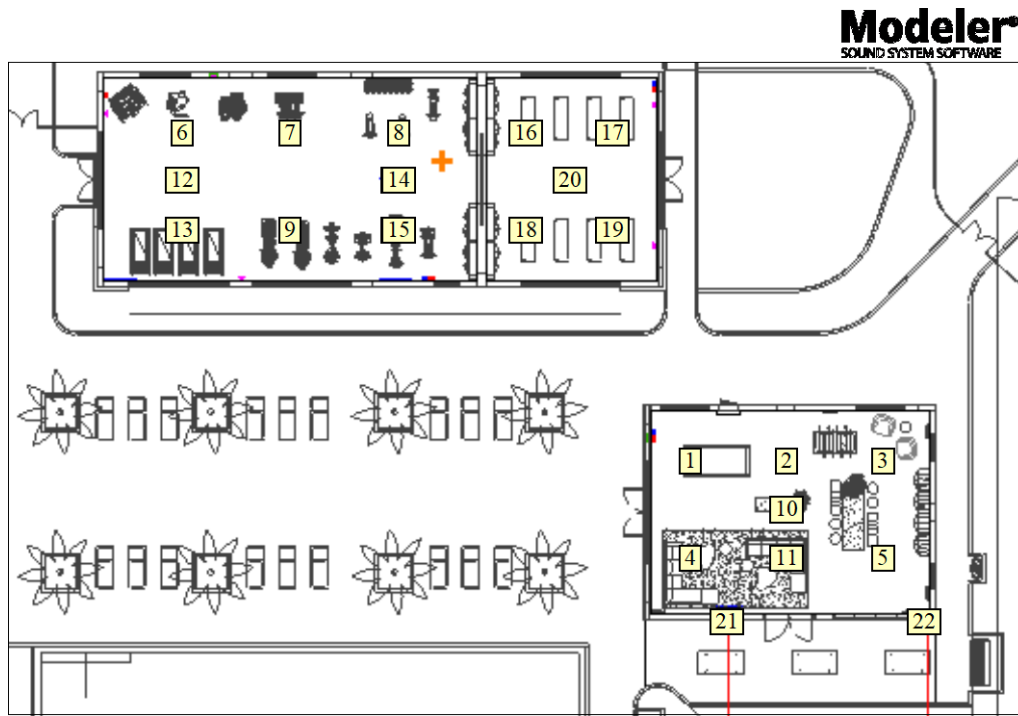
5.0 3D Modelling (COMMUNITY CENTER TH-02)

5.1 Electro-Acoustical room model

Based on the architectural drawings of the venue, we created a new electro-acoustical room model.



6.0 Loudspeaker Layout



6.1 Loudspeaker Database

Details																
	Cluster	Zone	Tier	Index	Description	Speaker	Status	Solo	X	Y	Z	Roll*	Pitch*	Yaw*	Pwr	Tap
1	1	1	1	1		Bose DM3C 100V	On	☐	11.620	-14.050	3.000	0.0	-90.0	0.0		25
2	2	1	1	1		Bose DM3C 100V	On	☐	16.120	-14.050	3.000	0.0	-90.0	0.0		25
3	3	1	1	1		Bose DM3C 100V	On	☐	20.620	-14.050	3.000	0.0	-90.0	0.0		25
4	4	1	1	1		Bose DM3C 100V	On	☐	11.620	-18.550	3.000	0.0	-90.0	0.0		25
5	5	1	1	1		Bose DM3C 100V	On	☐	20.620	-18.550	3.000	0.0	-90.0	0.0		25
6	6	1	1	1		Bose DM3C 100V	On	☐	-12.100	1.400	5.000	0.0	-90.0	0.0		25
7	7	1	1	1		Bose DM3C 100V	On	☐	-7.100	1.400	5.000	0.0	-90.0	0.0		25
8	8	1	1	1		Bose DM3C 100V	On	☐	-2.100	1.400	5.000	0.0	-90.0	0.0		25
9	9	1	1	1		Bose DM3C 100V	On	☐	-7.100	-3.100	5.000	0.0	-90.0	0.0		25
10	10	1	1	1		Bose DM8C-SUB 100V	On	☐	16.100	-16.300	3.000	0.0	-90.0	0.0		150
11	11	1	1	1		Bose DM3C 100V	On	☐	16.120	-18.550	3.000	0.0	-90.0	0.0		25
12	12	1	1	1		Bose DM8C-SUB 100V	On	☐	-12.100	-0.800	5.000	0.0	-90.0	0.0		150
13	13	1	1	1		Bose DM3C 100V	On	☐	-12.100	-3.100	5.000	0.0	-90.0	0.0		25
14	14	1	1	1		Bose DM8C-SUB 100V	On	☐	-2.100	-0.800	5.000	0.0	-90.0	0.0		150
15	15	1	1	1		Bose DM3C 100V	On	☐	-2.100	-3.100	5.000	0.0	-90.0	0.0		25
16	16	1	1	1		Bose DM3C 100V	On	☐	4.000	1.400	3.000	0.0	-90.0	0.0		25
17	17	1	1	1		Bose DM3C 100V	On	☐	8.000	1.400	3.000	0.0	-90.0	0.0		25
18	18	1	1	1		Bose DM3C 100V	On	☐	4.000	-3.100	3.000	0.0	-90.0	0.0		25
19	19	1	1	1		Bose DM3C 100V	On	☐	8.000	-3.100	3.000	0.0	-90.0	0.0		25
20	20	1	1	1		Bose DM8C-SUB 100V	On	☐	6.000	-0.800	3.000	0.0	-90.0	0.0		150
21	21	1	1	1		Bose DM3SE 100V	On	☐	13.300	-21.500	4.000	0.0	0.0	-90.0	25	
22	22	1	1	1		Bose DM3SE 100V	On	☐	22.500	-21.500	4.000	0.0	0.0	-90.0	25	

7.0 System Performance

7.1 Calculation Assumptions

- Power source: Broadband Pink Noise (6dB CF)
- Power settings (Loudspeakers): Maximum continuous (RMS) power ()
- Direct field coverage (Broadband, dBZ): Direct sound levels
- Listening Height: 1.5m from FFL

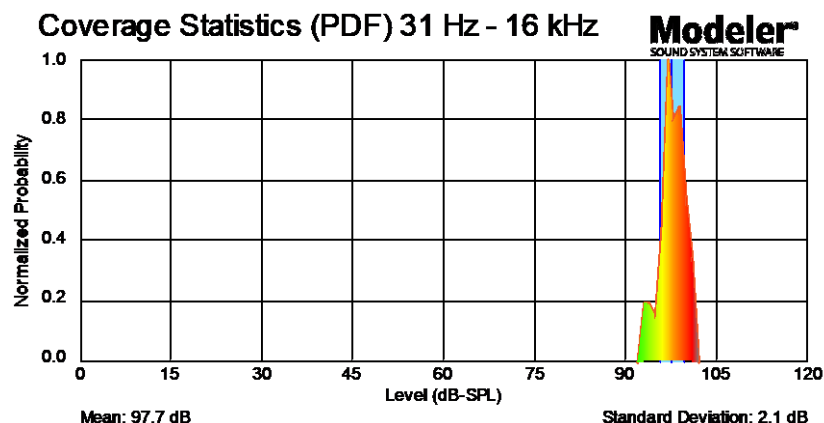
7.2 Direct field Coverage (31-16kHz) (GYM)

The first goal in creating a good sound system is to provide even direct sound intensity over the entire audience area and avoiding direct sound coverage in the non-occupied areas of the venue. The direct field is the sound that reaches the listener's ears directly from the speaker(s) in the design and it does not consider any sound reflections.

The coverage map/s below shows the direct field coverage for the venue. Good direct field coverage appears as relatively evenly coloured coverage area, without distinct dropout spots. This ensures even coverage of sound intensity over the coverage areas.



The information of this direct field (31-16kHz) coverage map can be condensed to a single figure of merit. The graph containing this figure, called a histogram, shows how often a given sound pressure level occurs in the model. Below you can find the SPL histogram of the direct field coverage between 31-16kHz. The frequency of occurrence (normalized) of a particular SPL is seen on the Y-axis of the graph while the X-axis shows the SPL in decibels.



A room with ideal coverage would have the same SPL at any point in the room and would appear as a single spike at that SPL level axis. As coverage becomes less than ideal the curve widens out to include other SPL levels.

The coverage data is processed in Modeler to yield the mean SPL level and the standard deviation. For the purposes of determining the quality or evenness of the coverage, it is the standard deviation that is important. (Applying more power to the system will increase the mean SPL level but the coverage pattern will be the same.)

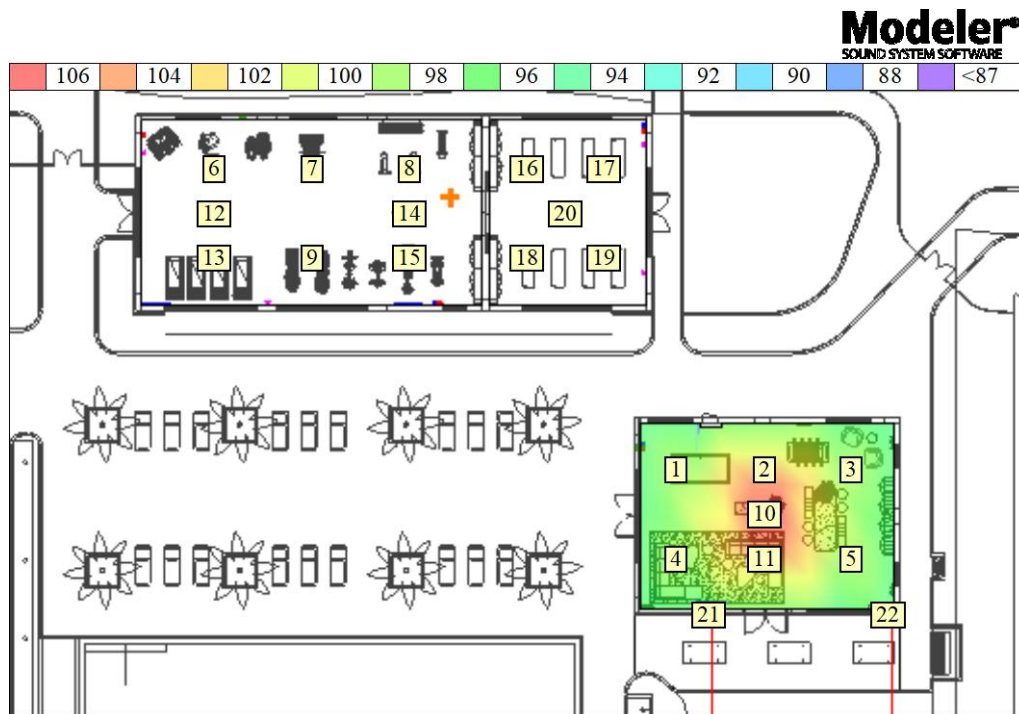
Referring to the standard practice of statistics, 68% of the samples are within +/- 1StDv and 95% of the samples are within +/- 2StDv.

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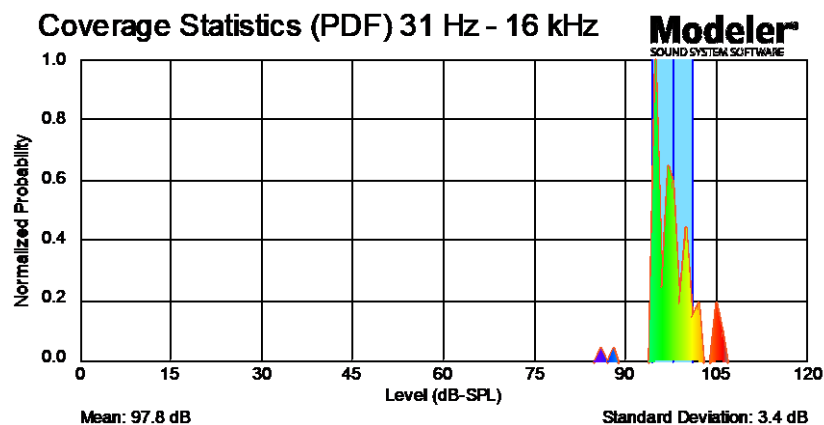
7.3 Direct field Coverage (31-16kHz) (COMMUNITY HALL)

The first goal in creating a good sound system is to provide even direct sound intensity over the entire audience area and avoiding direct sound coverage in the non-occupied areas of the venue. The direct field is the sound that reaches the listener's ears directly from the speaker(s) in the design and it does not consider any sound reflections.

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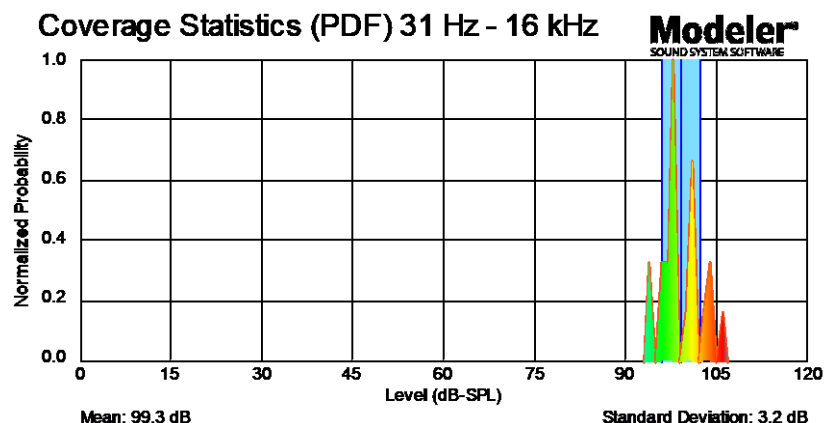
7.4 Direct field Coverage (31-16kHz) (YOGA)

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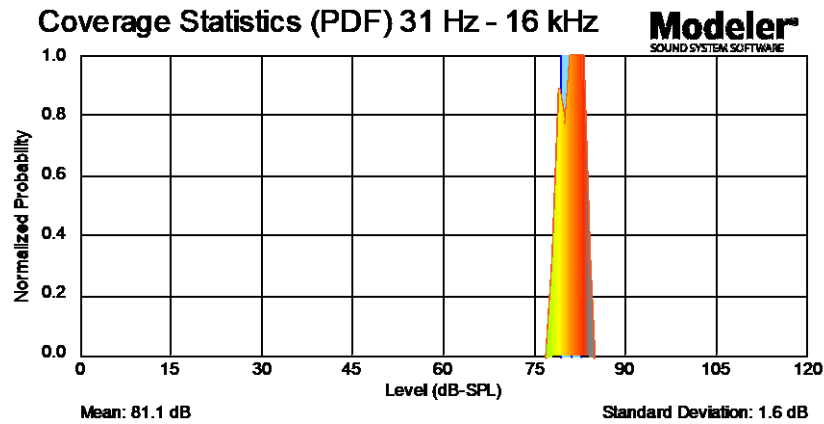
7.5 Direct field Coverage (31-16kHz) (OUTDOOR AREA)

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A room with ideal coverage would have the same SPL at any point in the room and would appear as a single spike at that SPL level axis. As coverage becomes less than ideal the curve widens out to include other SPL levels.

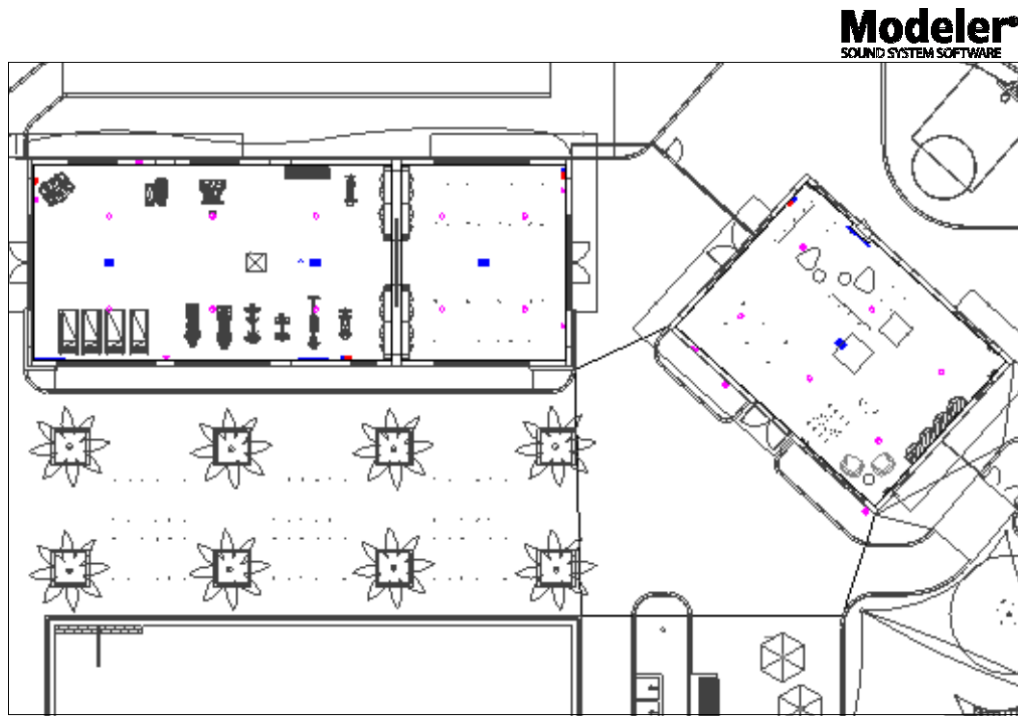
The coverage data is processed in Modeler to yield the mean SPL level and the standard deviation. For the purposes of determining the quality or evenness of the coverage, it is the standard deviation that is important. (Applying more power to the system will increase the mean SPL level but the coverage pattern will be the same.)

Referring to the standard practice of statistics, 68% of the samples are within $\pm 1\text{StDv}$ and 95% of the samples are within $\pm 2\text{StDv}$.

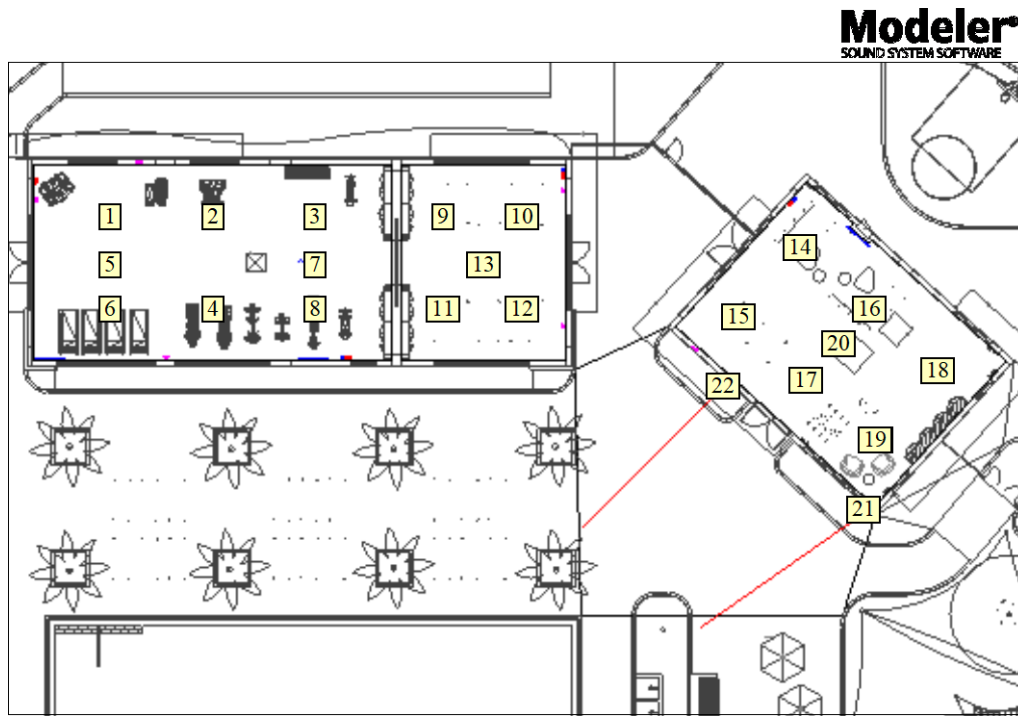
8.0 3D Modelling (COMMUNITY CENTER TH-03)

8.1 Electro-Acoustical room model

Based on the architectural drawings of the venue, we created a new electro-acoustical room model.



9.0 Loudspeaker Layout



9.1 Loudspeaker Database

Details																	
	Cluster	Zone	Tier	Index	Description	Speaker	Status	Solo	X	Y	Z	Roll°	Pitch°	Yaw°	Pwr	Tap	
1	1	1	1	1	▼	Bose DM3C 100V	▼	On	□	31.000	43.900	5.000	0.0	-90.0	0.0	25	▼
2	2	1	1	1	▼	Bose DM3C 100V	▼	On	□	36.000	43.900	5.000	0.0	-90.0	0.0	25	▼
3	3	1	1	1	▼	Bose DM3C 100V	▼	On	□	41.000	43.900	5.000	0.0	-90.0	0.0	25	▼
4	4	1	1	1	▼	Bose DM3C 100V	▼	On	□	36.000	39.400	5.000	0.0	-90.0	0.0	25	▼
5	5	1	1	1	▼	Bose DM8C-SUB 100V	▼	On	□	31.000	41.600	5.000	0.0	-90.0	0.0	150	▼
6	6	1	1	1	▼	Bose DM3C 100V	▼	On	□	31.000	39.400	5.000	0.0	-90.0	0.0	25	▼
7	7	1	1	1	▼	Bose DM8C-SUB 100V	▼	On	□	41.000	41.600	5.000	0.0	-90.0	0.0	150	▼
8	8	1	1	1	▼	Bose DM3C 100V	▼	On	□	41.000	39.400	5.000	0.0	-90.0	0.0	25	▼
9	9	1	1	1	▼	Bose DM3C 100V	▼	On	□	47.100	43.900	3.000	0.0	-90.0	0.0	25	▼
10	10	1	1	1	▼	Bose DM3C 100V	▼	On	□	51.100	43.900	3.000	0.0	-90.0	0.0	25	▼
11	11	1	1	1	▼	Bose DM3C 100V	▼	On	□	47.100	39.400	3.000	0.0	-90.0	0.0	25	▼
12	12	1	1	1	▼	Bose DM3C 100V	▼	On	□	51.100	39.400	3.000	0.0	-90.0	0.0	25	▼
13	13	1	1	1	▼	Bose DM8C-SUB 100V	▼	On	□	49.100	41.600	3.000	0.0	-90.0	0.0	150	▼
14	14	1	1	1	▼	Bose DM3C 100V	▼	On	□	64.500	42.400	3.000	0.0	-90.0	0.0	25	▼
15	15	1	1	1	▼	Bose DM3C 100V	▼	On	□	61.500	39.100	3.000	0.0	-90.0	0.0	25	▼
16	16	1	1	1	▼	Bose DM3C 100V	▼	On	□	67.900	39.400	3.000	0.0	-90.0	0.0	25	▼
17	17	1	1	1	▼	Bose DM3C 100V	▼	On	□	64.800	36.000	3.000	0.0	-90.0	0.0	25	▼
18	18	1	1	1	▼	Bose DM3C 100V	▼	On	□	71.200	36.400	3.000	0.0	-90.0	0.0	25	▼
19	19	1	1	1	▼	Bose DM3C 100V	▼	On	□	68.200	33.000	3.000	0.0	-90.0	0.0	25	▼
20	20	1	1	1	▼	Bose DM8C-SUB 100V	▼	On	□	66.400	37.700	3.000	0.0	-90.0	0.0	150	▼
21	21	1	1	1	▼	Bose DM3SE 100V	▼	On	□	67.600	29.600	4.000	90.0	-10.0	-145.0	25	▼
22	22	1	1	1	▼	Bose DM3SE 100V	▼	On	□	60.800	35.700	4.000	90.0	-10.0	-135.0	25	▼

10.0 System Performance

10.1 Calculation Assumptions

- Power source: Broadband Pink Noise (6dB CF)
- Power settings (Loudspeakers): Maximum continuous (RMS) power ()
- Direct field coverage (Broadband, dBZ): Direct sound levels
- Listening Height: 1.5m from FFL

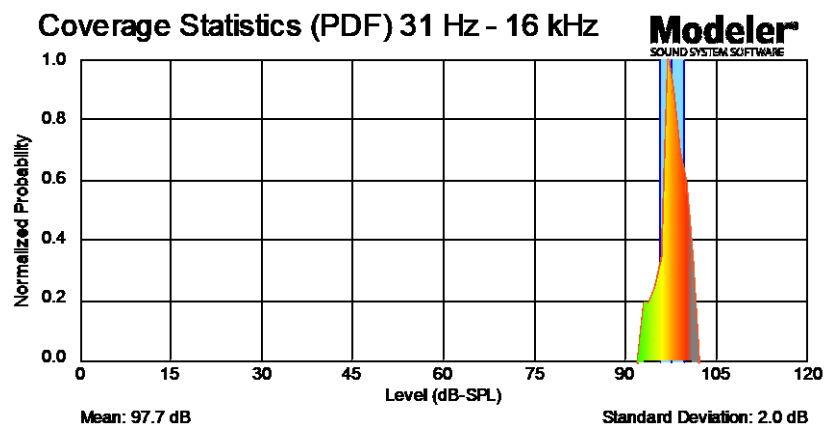
10.2 Direct field Coverage (31-16kHz) (GYM)

The first goal in creating a good sound system is to provide even direct sound intensity over the entire audience area and avoiding direct sound coverage in the non-occupied areas of the venue. The direct field is the sound that reaches the listener's ears directly from the speaker(s) in the design and it does not consider any sound reflections.

The coverage map/s below shows the direct field coverage for the venue. Good direct field coverage appears as relatively evenly coloured coverage area, without distinct dropout spots. This ensures even coverage of sound intensity over the coverage areas.



The information of this direct field (31-16kHz) coverage map can be condensed to a single figure of merit. The graph containing this figure, called a histogram, shows how often a given sound pressure level occurs in the model. Below you can find the SPL histogram of the direct field coverage between 31-16kHz. The frequency of occurrence (normalized) of a particular SPL is seen on the Y-axis of the graph while the X-axis shows the SPL in decibels.



A room with ideal coverage would have the same SPL at any point in the room and would appear as a single spike at that SPL level axis. As coverage becomes less than ideal the curve widens out to include other SPL levels.

The coverage data is processed in Modeler to yield the mean SPL level and the standard deviation. For the purposes of determining the quality or evenness of the coverage, it is the standard deviation that is important. (Applying more power to the system will increase the mean SPL level but the coverage pattern will be the same.)

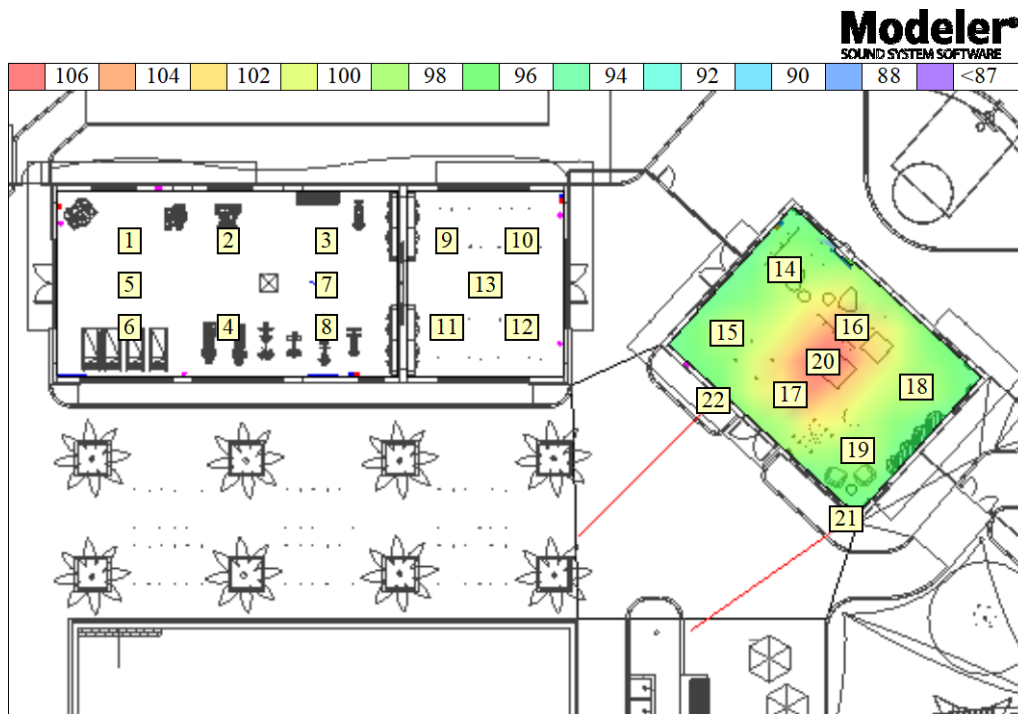
Referring to the standard practice of statistics, 68% of the samples are within +/- 1StDv and 95% of the samples are within +/- 2StDv.

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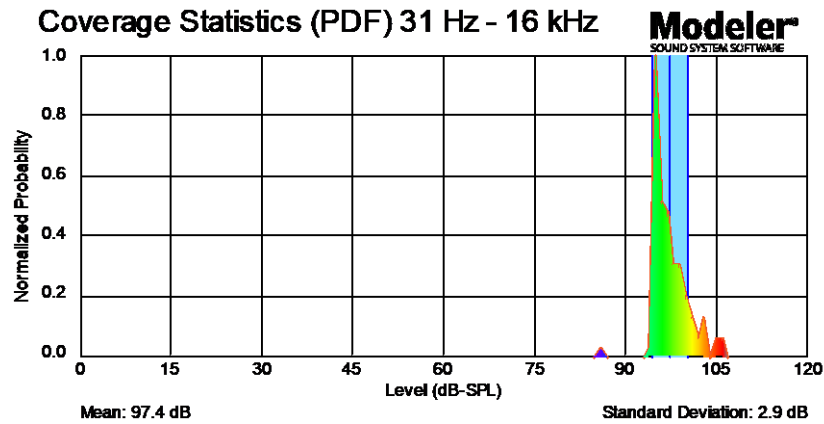
10.3 Direct field Coverage (31-16kHz) (COMMUNITY HALL)

The first goal in creating a good sound system is to provide even direct sound intensity over the entire audience area and avoiding direct sound coverage in the non-occupied areas of the venue. The direct field is the sound that reaches the listener's ears directly from the speaker(s) in the design and it does not consider any sound reflections.

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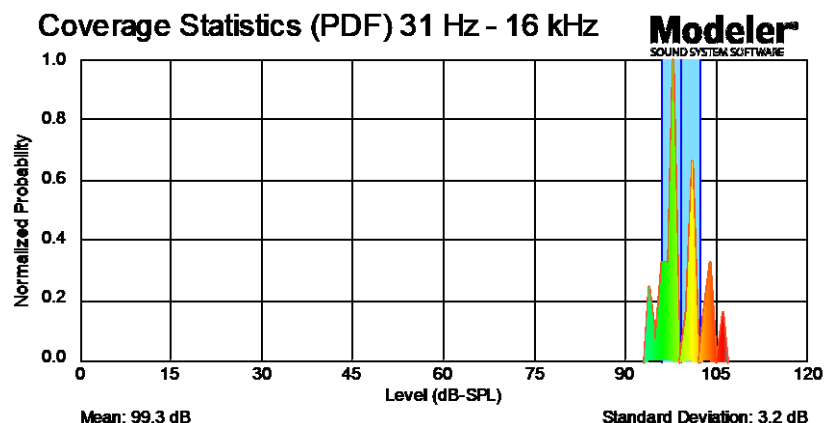
10.4 Direct field Coverage (31-16kHz) (YOGA)

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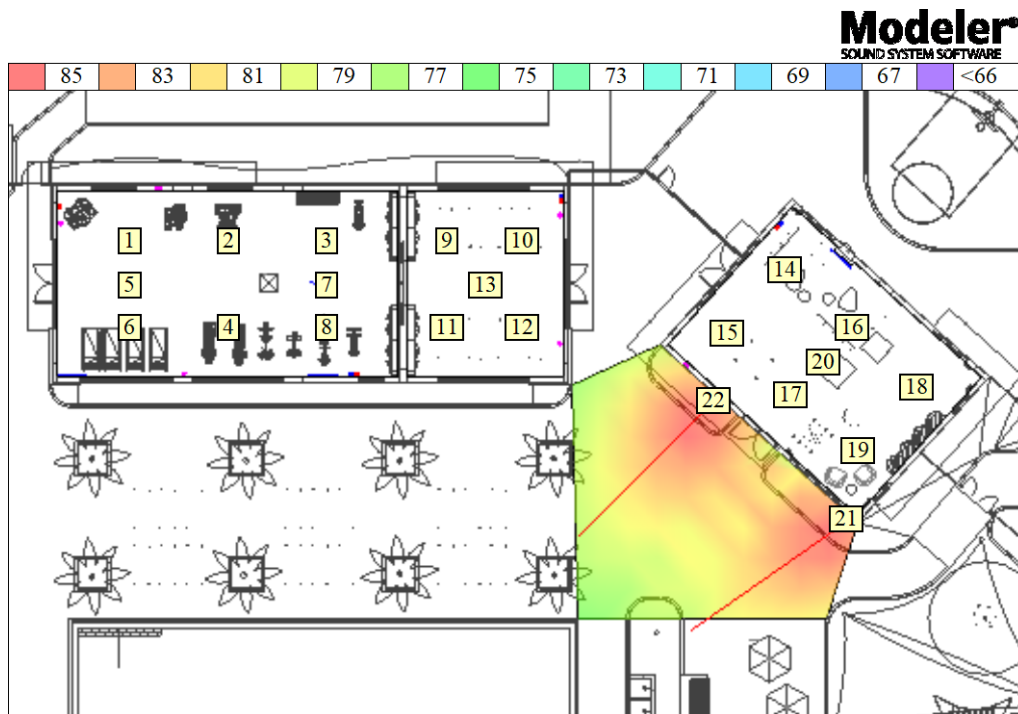
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Referring to the standard practice of statistics, 68% of the samples are within $\pm 1\text{StDv}$ and 95% of the samples are within $\pm 2\text{StDv}$.

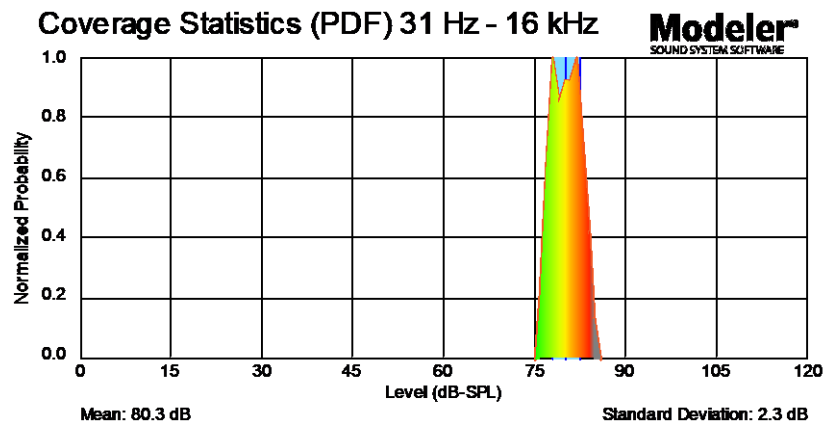
10.5 Direct field Coverage (31-16kHz) (OUTDOOR AREA)

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